House Price Predictor Model

Objective:

Build a House Price Prediciton Model For Houses in Karachi

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
path = 'Entities.csv'
df = pd.read_csv(path)
df.head()
```

Out[2]:

	Unnamed: 0	property_id	location_id	page_url	property_type	price	location	city	province_name	latitude	longitude	baths	purpose	bedrooms	date_added	agency	a
0	0	237062	3325	https://www.zameen.com/Property/g_10_g_10_2_gr...	Flat	10000000	G-10	Islamabad	Islamabad Capital	33.679890	73.012640	2	For Sale	2	2/4/2019	NaN	
1	1	346905	3236	https://www.zameen.com/Property/e_11_2_service...	Flat	6900000	E-11	Islamabad	Islamabad Capital	33.700993	72.971492	3	For Sale	3	5/4/2019	NaN	
2	2	386513	764	https://www.zameen.com/Property/islamabad_g_15...	House	16500000	G-15	Islamabad	Islamabad Capital	33.631486	72.926559	6	For Sale	5	7/17/2019	NaN	
3	3	656161	340	https://www.zameen.com/Property/islamabad_bani...	House	43500000	Bani Gala	Islamabad	Islamabad Capital	33.707573	73.151199	4	For Sale	4	4/5/2019	NaN	
4	4	841645	3226	https://www.zameen.com/Property/dha_valley_dha...	House	7000000	DHA Defence	Islamabad	Islamabad Capital	33.492591	73.301339	3	For Sale	3	7/10/2019	Easy Property	Muham Junaid Muham Sh Dir

removing unwanted columns

In [3]:
```
df.columns
```

Out[3]:
```
Index(['Unnamed: 0', 'property_id', 'location_id', 'page_url', 'property_type',
       'price', 'location', 'city', 'province_name', 'latitude', 'longitude',
       'baths', 'purpose', 'bedrooms', 'date_added', 'agency', 'agent',
       'Total_Area'],
      dtype='object')
```

In [4]:
```
cols_to_keep = ['property_type','purpose','baths','bedrooms','location','Total_Area','city','price','page_url']
df =  df[cols_to_keep]
```

In [5]:
```
cols_rename = {'Total_Area': 'sqft'}
df.rename(columns=cols_rename,inplace=True)
```

In [6]:
```
#subsetting for karachi
df = df[df['city']=='Karachi']
```

In [7]:
```
df.drop(columns=['city'],inplace=True)
```

In [8]:
```
df['price'] = df['price']/1e5
```

In [9]:
```
df['property_type'].unique()
```

Out[9]:
```
array(['House', 'Flat', 'Room', 'Lower Portion', 'Upper Portion',
       'Penthouse', 'Farm House'], dtype=object)
```

In [10]:
```
# keeping the model genereic. removin penthhouse and farm house
property_to_rem = ['Penthouse','Farm House','Room']
bool_mask = df['property_type'].isin(property_to_rem)
df = df[~(bool_mask)]
df.property_type.unique()
```

Out[10]:
```
array(['House', 'Flat', 'Lower Portion', 'Upper Portion'], dtype=object)
```

In [11]:
```
#imputing values for bathrooms and bedrooms where the value is 0 with its median value
bath_med = df['baths'].median()
bed_med = df['bedrooms'].median()

df.loc[df['baths'] == 0, 'baths'] = bath_med
df.loc[df['bedrooms'] == 0, 'bedrooms'] = bed_med
```

In [12]:
```
df[(df['baths'] == 0) | (df['bedrooms'] == 0)]
```

Out[12]:

property_type	purpose	baths	bedrooms	location	sqft	price	page_url

In [13]:
```
df.shape
```

Out[13]:
```
(59960, 8)
```

In [14]:
```
mask = df['baths']>df['bedrooms']+2
df[mask].head()
```

Out[14]:

	property_type	purpose	baths	bedrooms	location	sqft	price	page_url
1738	House	For Sale	7	4	Scheme 33	13068.048	115.0	https://www.zameen.com/Property/scheme_33_hadi...
2068	House	For Sale	9	6	Gulshan-e-Iqbal Town	98010.000	950.0	https://www.zameen.com/Property/gulshan_e_iqba...
2279	House	For Rent	8	5	Jamshed Town	4356.016	3.0	https://www.zameen.com/Property/pechs_pechs_bl...
3162	House	For Sale	7	4	Korangi	13068.048	160.0	https://www.zameen.com/Property/korangi_mbchs_...
3775	House	For Sale	10	6	Cantt	59895.000	950.0	https://www.zameen.com/Property/malir_cantonme...

In [15]:
```
bckp_df = df.copy(deep=True) # to keep backup
```

In [16]:
```
# removing unsual number of beds and bathrooms

mask1 = df['baths'] > df['bedrooms'] + 2  # Baths exceed bedrooms by more than 2
mask2 = df['baths'] < df['bedrooms'] / 2   # Baths are less than half the number of bedrooms

df = df[~(mask1 | mask2)]
```

AREA BASED ANAMOLY FILTERING

In [17]:
```
df['sqft'].describe()
```

Out[17]:
```
count    5.924800e+04
mean     1.329492e+04
std      2.118058e+04
min      0.000000e+00
25%      3.267012e+03
50%      1.143454e+04
75%      1.579056e+04
max      3.267000e+06
Name: sqft, dtype: float64
```

In [18]:
```
df = df[~(df['sqft'] == 0)]
```

In [19]:
```
df['sqft'].describe()
```

Out[19]:
```
count    5.924700e+04
mean     1.329514e+04
std      2.118068e+04
min      2.722510e+02
25%      3.267012e+03
50%      1.143454e+04
75%      1.579056e+04
max      3.267000e+06
Name: sqft, dtype: float64
```

In [20]:
```
mask = df['sqft'] == df['sqft'].quantile(0.75)
df[mask].head()
```

Out[20]:

	property_type	purpose	baths	bedrooms	location	sqft	price	page_url
540	Flat	For Sale	2	2	Fazaia Housing Scheme	15790.558	46.66	https://www.zameen.com/Property/karachi_fazaia...
728	Flat	For Sale	2	2	Fazaia Housing Scheme	15790.558	46.66	https://www.zameen.com/Property/karachi_fazaia...
849	Flat	For Sale	2	2	Fazaia Housing Scheme	15790.558	46.66	https://www.zameen.com/Property/karachi_fazaia...
857	Flat	For Sale	2	2	Fazaia Housing Scheme	15790.558	46.66	https://www.zameen.com/Property/karachi_fazaia...
1015	Flat	For Sale	2	2	Fazaia Housing Scheme	15790.558	46.66	https://www.zameen.com/Property/karachi_fazaia...

In [21]:
```
x = df.loc[540]
print(x)
x['sqft']/(x['bedrooms'])
```

```
property_type                                               Flat
purpose                                                 For Sale
baths                                                          2
bedrooms                                                       2
location                                   Fazaia Housing Scheme
sqft                                                   15790.558
price                                                      46.66
page_url         https://www.zameen.com/Property/karachi_fazaia...
Name: 540, dtype: object
```

Out[21]:
```
7895.279
```

having 15000 sqft is unusual for 2 baths bed and 3bath and bed. It is an anamoly

In [22]:
```
bckp_df = df.copy(deep=True) # to keep backup
```

In [23]:
```
df.shape
```

Out[23]:
```
(59247, 8)
```

In [24]:
```
df[df['sqft']>10000]
```

Out[24]:

	property_type	purpose	baths	bedrooms	location	sqft	price	page_url
156	House	For Sale	7	6	Cantt	21780.000	4500.0	https://www.zameen.com/Property/faisal_cantonm...
158	Flat	For Sale	3	3	DHA Defence	24230.339	210.0	https://www.zameen.com/Property/d_h_a_dha_phas...
160	House	For Sale	4	4	Gadap Town	26136.096	130.0	https://www.zameen.com/Property/surjani_town_s...
163	Flat	For Sale	4	4	Gulshan-e-Iqbal Town	26680.598	165.0	https://www.zameen.com/Property/gulshan_e_iqba...
165	House	For Sale	6	5	Scheme 33	26136.096	128.0	https://www.zameen.com/Property/scheme_33_saad...
...	...	...	...	...	...	...	...	...
168440	Flat	For Sale	3	2	Gadap Town	10345.538	48.0	https://www.zameen.com/Property/gulshan_e_maym...
168441	House	For Sale	3	6	Gadap Town	26136.096	265.0	https://www.zameen.com/Property/gulshan_e_maym...
168443	House	For Sale	3	6	Gadap Town	26136.096	270.0	https://www.zameen.com/Property/gulshan_e_maym...
168444	House	For Sale	3	3	Gadap Town	21235.578	110.0	https://www.zameen.com/Property/gulshan_e_maym...
168445	House	For Sale	3	3	Bahria Town Karachi	25591.594	90.0	https://www.zameen.com/Property/bahria_town_ka...

31475 rows × 8 columns

area data is very inaccurate. In order to remedy this we will use the page url to scrap the correct area and convert it back to sqft

In [25]:
```
df.reset_index(drop=True,inplace=True)
```

In [26]:
```
# df.to_csv('output.csv', index=False, sep=',', encoding='utf-8')
```

In [27]:
```
ktown = pd.read_csv('corrected_areas.csv')
```

In [28]:
```
ktown.head()
```

Out[28]:

	property_type	purpose	baths	bedrooms	location	sqft	price	page_url	area	val	unit	area_sqft
0	House	For Sale	7	6	Cantt	21780.000	4500.0	https://www.zameen.com/Property/faisal_cantonm...	2,000 Sq. Yd.	2,000	Sq. Yd.	18000.0
1	House	For Sale	8	6	Gulistan-e-Jauhar	4356.016	350.0	https://www.zameen.com/Property/karachi_gulist...	400 Sq. Yd.	400	Sq. Yd.	3600.0
2	Flat	For Sale	3	3	DHA Defence	24230.339	210.0	https://www.zameen.com/Property/d_h_a_dha_phas...	222 Sq. Yd.	222	Sq. Yd.	1998.0
3	House	For Sale	1	2	Malir	8712.032	65.0	https://www.zameen.com/Property/malir_malir_ka...	106 Sq. Yd.	106	Sq. Yd.	954.0
4	House	For Sale	4	4	Gadap Town	26136.096	130.0	https://www.zameen.com/Property/surjani_town_s...	240 Sq. Yd.	240	Sq. Yd.	2160.0

In [29]:
```
ktown.columns
```

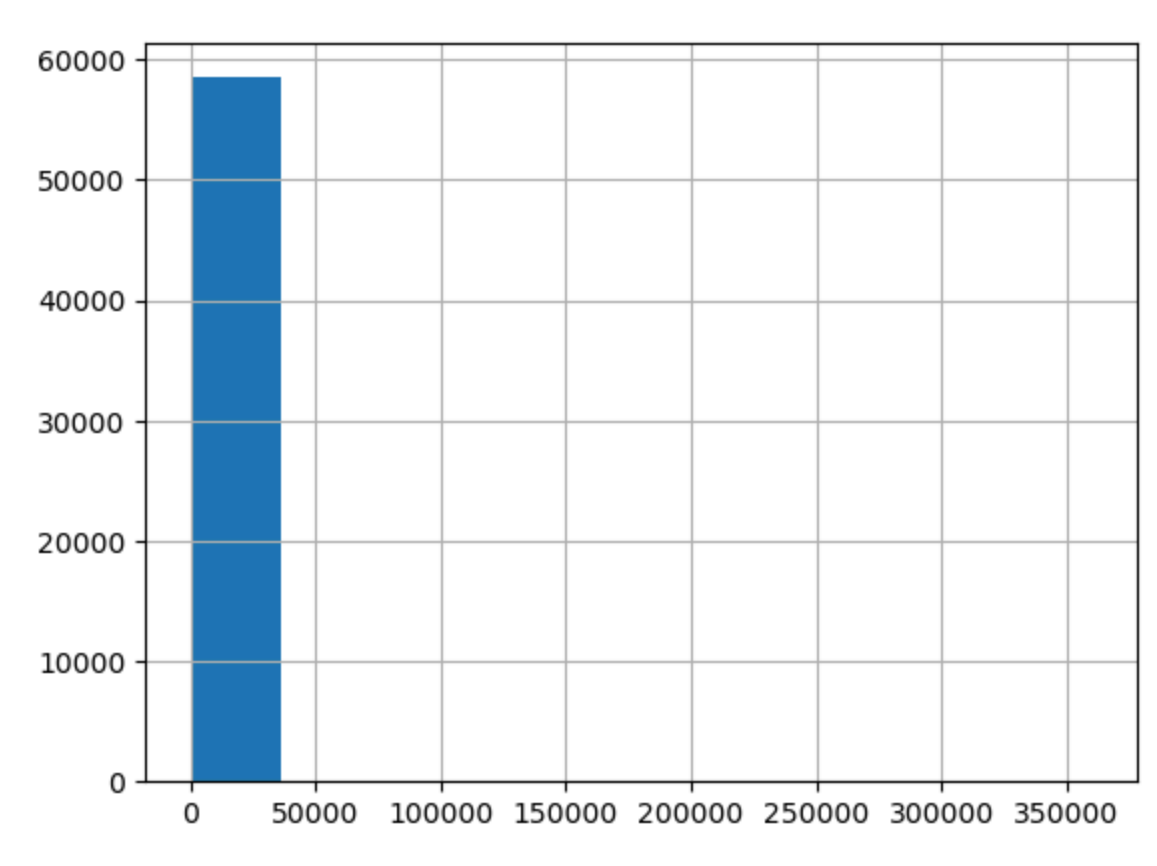
Out[29]:
```
Index(['property_type', 'purpose', 'baths', 'bedrooms', 'location', 'sqft',
       'price', 'page_url', 'area', 'val', 'unit', 'area_sqft'],
      dtype='object')
```

In [30]:
```
cols_to_keep = ['property_type', 'purpose', 'baths', 'bedrooms', 'location', 'area_sqft',
       'price']

ktown = ktown[cols_to_keep]
```

In [31]:
```
ktown['area_sqft'].hist()
```

Out[31]:
```
<Axes: >
```



In [32]:
```
ktown['area_sqft'].max()
```

Out[32]:
```
360000.0
```

In [33]:
```
ktown['area_sqft'].describe()
```

Out[33]:
```
count     58519.000000
mean       2021.419897
std        2905.928377
min          54.000000
25%         999.000000
50%        1404.000000
75%        2241.000000
max      360000.000000
Name: area_sqft, dtype: float64
```

In [34]:
```
#removing outlies using iqr method

q1 = ktown['area_sqft'].quantile(0.25)
q3 = ktown['area_sqft'].quantile(0.75)

iqr = q3-q1

lb = q1 - 1.5 * iqr #lower bound
ub = q3 + 1.5 * iqr #upper bound

lb , ub
```

Out[34]:
```
(-864.0, 4104.0)
```

In [35]:
```
mask = (ktown['area_sqft']>=lb) & (ktown['area_sqft']<=ub)

ktown = ktown[mask]
```

In [36]:
```
ktown['area_sqft'].describe()
```

Out[36]:
```
count    51882.000000
mean      1495.948729
std        752.342129
min         54.000000
25%        954.000000
50%       1251.000000
75%       1800.000000
max       4104.000000
Name: area_sqft, dtype: float64
```

In [37]:
```
# finding unsually very tight or impractical houses.

mask = ktown['area_sqft']/ktown['bedrooms']<300

ktown = ktown[~mask]
```

finding unsually spacious places

In [38]:
```
ktown[ktown['area_sqft']/ktown['bedrooms']>1000]
```

Out[38]:

	property_type	purpose	baths	bedrooms	location	area_sqft	price
44	Flat	For Sale	3	2	DHA Defence	2052.0	350.00
45	Flat	For Sale	1	1	DHA Defence	1350.0	250.00
56	Lower Portion	For Rent	3	3	Gulshan-e-Iqbal Town	3600.0	1.50
57	Flat	For Sale	1	1	DHA Defence	1242.0	305.00
112	House	For Sale	4	3	Gulshan-e-Iqbal Town	3600.0	400.00
...	...	...	...	...	...	...	...
58356	House	For Sale	3	3	Bahria Town Karachi	3150.0	156.00
58366	House	For Sale	3	3	Bahria Town Karachi	3150.0	145.00
58417	Lower Portion	For Rent	4	4	Gulistan-e-Jauhar	4050.0	0.45
58443	Upper Portion	For Rent	3	3	Gulistan-e-Jauhar	4050.0	0.45
58466	House	For Sale	3	3	North Nazimabad	3969.0	475.00

1168 rows × 7 columns

In [39]:
```
ktown[ktown['area_sqft']/ktown['bedrooms']>1000]['bedrooms'].mean()
```

Out[39]:
```
2.720890410958904
```

the mean represent that the area/bed above 1000 is unusual and potentially an anomaly

In [40]:
```
mask = ktown['area_sqft']/ktown['bedrooms']>1000

ktown = ktown[~mask]
```

In [41]:
```
bckp_df = ktown.copy(deep=True)
```

filtering outliers regarding prices

dataset contain two sets of pricing: for sale and for rent. We need to find outliers based on purpose as well as the location

In [42]:
```
ktown['price_per_sqft'] = ktown['price']*1e5/ktown['area_sqft']
```

In [43]:
```
ktown['price'] = ktown['price'] * 1e5
ktown_rent = ktown[ktown['purpose']=='For Rent'].copy(deep=True)  # rent
ktown_sale = ktown[ktown['purpose']=='For Sale'].copy(deep=True)  # sale
```

In [44]:
```
def remove_outliers_by_location(df):
    cleaned_data = []
    for location, group in df.groupby('location'):
        Q1 = group['price_per_sqft'].quantile(0.25)
        Q3 = group['price_per_sqft'].quantile(0.75)
        IQR = Q3 - Q1

        lower_bound = Q1 - 1.5 * IQR
        upper_bound = Q3 + 1.5 * IQR

        cleaned_group = group[(group['price_per_sqft'] >= lower_bound) & (group['price_per_sqft'] <= upper_bound)]
        cleaned_data.append(cleaned_group)

    return pd.concat(cleaned_data)
```

In [45]:
```
ktown_sale_cleaned = remove_outliers_by_location(ktown_sale)
```

In [46]:
```
ktown_rent_cleaned = remove_outliers_by_location(ktown_rent)
```

In [47]:
```
ktown_cleaned = pd.concat([ktown_sale_cleaned, ktown_rent_cleaned ], ignore_index=True)
```

In [48]:
```
ktown_cleaned.drop(columns=['price_per_sqft'],inplace=True)
```

In [49]:
```
ktown_cleaned['price'] = ktown_cleaned['price']/1e5
```

In [50]:
```
ktown_cleaned
```

Out[50]:

	property_type	purpose	baths	bedrooms	location	area_sqft	price
0	House	For Sale	6	8	APP Employees Co-operative Housing Society	3600.0	390.00
1	House	For Sale	9	8	APP Employees Co-operative Housing Society	3600.0	390.00
2	Flat	For Sale	2	2	ASF Housing Scheme	846.0	30.56
3	Flat	For Sale	2	2	ASF Housing Scheme	1152.0	28.00
4	Flat	For Sale	3	3	ASF Tower	1575.0	85.00
...	...	...	...	...	...	...	...
41537	House	For Rent	5	5	Zamzama	3150.0	2.80
41538	House	For Rent	5	5	Zamzama	3150.0	3.25
41539	House	For Rent	4	4	Zamzama	3150.0	2.75
41540	House	For Rent	4	4	Zamzama	3150.0	2.40
41541	House	For Rent	4	5	Zamzama	3150.0	2.50

41542 rows × 7 columns

DATA PREPPING FOR TRAINING

In [51]:
```
ktown_prcsd= ktown_cleaned.copy(deep=True)
```

In [52]:
```
locations = ktown_cleaned['location'].unique()
prop_type = ktown_cleaned['property_type'].unique()
purpose = ktown_cleaned['purpose'].unique()
```

In [53]:
```
# l = len(ktown_cleaned['purpose'].unique())

# purpose_map = dict(zip(ktown_cleaned['purpose'].unique(),np.arange(l)))

# l = len(ktown_cleaned['property_type'].unique())
# property_type_map = dict(zip(ktown_cleaned['property_type'].unique(),np.arange(l)))

# print(purpose_map)
# print(property_type_map)

# ktown_cleaned['purpose'] = ktown_cleaned['purpose'].map(purpose_map)

# ktown_cleaned['property_type'].map(property_type_map)
```

In [54]:
```
ktown_prcsd.head(3)
```

Out[54]:

	property_type	purpose	baths	bedrooms	location	area_sqft	price
0	House	For Sale	6	8	APP Employees Co-operative Housing Society	3600.0	390.00
1	House	For Sale	9	8	APP Employees Co-operative Housing Society	3600.0	390.00
2	Flat	For Sale	2	2	ASF Housing Scheme	846.0	30.56

In [55]:
```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

In [56]:
```
# Define the target variable and features
y = ktown_prcsd['price']
X = ktown_prcsd.drop(columns=['price'])

# Define categorical and numerical columns
categorical_cols = ['property_type', 'purpose', 'location']
numerical_cols = ['baths', 'bedrooms', 'area_sqft']

# Scale numerical features
scaler = StandardScaler()
X[numerical_cols] = scaler.fit_transform(X[numerical_cols])

# One-hot encode categorical features
encoder = OneHotEncoder(sparse_output=False, drop='first')  # drop='first' to avoid dummy variable trap
encoded_cats = encoder.fit_transform(X[categorical_cols])

# Create a DataFrame for the encoded categorical features
encoded_cats_df = pd.DataFrame(encoded_cats, columns=encoder.get_feature_names_out(categorical_cols))

# Concatenate the numerical and encoded categorical DataFrames
X_processed = pd.concat([X[numerical_cols], encoded_cats_df], axis=1)

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X_processed, y, test_size=0.2, random_state=42)

# Display the shapes of the training and testing sets
print(f'Training set shape: {X_train.shape}, {y_train.shape}')
print(f'Testing set shape: {X_test.shape}, {y_test.shape}')
```

```
Training set shape: (33233, 197), (33233,)
Testing set shape: (8309, 197), (8309,)
```

In [57]:
```
X_processed.head()
```

Out[57]:

	baths	bedrooms	area_sqft	property_type_House	property_type_Lower Portion	property_type_Upper Portion	purpose_For Sale	location_ASF Housing Scheme	location_ASF Tower	location_Abdullah Haroon Road	...	location_Super Highway	location_Surti Muslim Co-Operative Housing Society	location_Tariq Road	location_Teacher Society
0	3.173237	5.247210	2.960255	1.0	0.0	0.0	1.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0
1	6.283434	5.247210	2.960255	1.0	0.0	0.0	1.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0
2	-0.973693	-0.910261	-1.007199	0.0	0.0	0.0	1.0	1.0	0.0	0.0	...	0.0	0.0	0.0	0.0
3	-0.973693	-0.910261	-0.566371	0.0	0.0	0.0	1.0	1.0	0.0	0.0	...	0.0	0.0	0.0	0.0
4	0.063039	0.115984	0.043009	0.0	0.0	0.0	1.0	0.0	1.0	0.0	...	0.0	0.0	0.0	0.0

5 rows × 197 columns

Model Selection

In [58]:
```
from sklearn.model_selection import GridSearchCV
from sklearn.linear_model import LinearRegression, Lasso, Ridge
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
```

In [61]:
```
# Define the models and their hyperparameters
models = {
    'Linear Regression': LinearRegression(),
    'Lasso Regression': Lasso(),
    'Ridge Regression': Ridge(),
    'Decision Tree': DecisionTreeRegressor(),
    'Random Forest': RandomForestRegressor()
}

# Define the hyperparameters for each model
param_grid = {
    'Linear Regression': {},
    'Lasso Regression': {'alpha': [0.01, 0.1, 1, 10, 100]},
    'Ridge Regression': {'alpha': [0.01, 0.1, 1, 10, 100]},
    'Decision Tree': {'max_depth': [None, 5, 10, 15, 20], 'min_samples_split': [2, 5, 10]},
    'Random Forest': {'n_estimators': [50, 100, 200], 'max_depth': [5, 10, 15], 'min_samples_split': [2, 5, 10]}
}
```

In [62]:
```
# # Initialize a dictionary to hold the best models
# best_models = {}

# # Loop through each model and perform GridSearchCV
# for model_name, model in models.items():
#     grid_search = GridSearchCV(estimator=model, param_grid=param_grid[model_name],
#                                scoring=['r2','neg_mean_squared_error'],refit='r2', cv=5, n_jobs=-1, verbose=1)
#     grid_search.fit(X_train, y_train)

#     # Store the best model and its parameters
#     best_models[model_name] = {
#         'best_model': grid_search.best_estimator_,
#         'best_params': grid_search.best_params_,
#         'best_score': grid_search.best_score_
#     }

# # Output the best models and their parameters
# for model_name, details in best_models.items():
#     print(f"{model_name}:")
#     print(f"  Best Parameters: {details['best_params']}")
#     print(f"  Best Score: {details['best_score']}")
```

```
Fitting 5 folds for each of 1 candidates, totalling 5 fits
Fitting 5 folds for each of 5 candidates, totalling 25 fits
Fitting 5 folds for each of 5 candidates, totalling 25 fits
Fitting 5 folds for each of 15 candidates, totalling 75 fits
Fitting 5 folds for each of 27 candidates, totalling 135 fits
Linear Regression:
  Best Parameters: {}
  Best Score: -4.299840000852746e+19
Lasso Regression:
  Best Parameters: {'alpha': 0.01}
  Best Score: 0.6817133978338602
Ridge Regression:
  Best Parameters: {'alpha': 1}
  Best Score: 0.6828938126728473
Decision Tree:
  Best Parameters: {'max_depth': None, 'min_samples_split': 10}
  Best Score: 0.9160457936334401
Random Forest:
  Best Parameters: {'max_depth': 15, 'min_samples_split': 10, 'n_estimators': 50}
  Best Score: 0.9138742812379153
```

In [ ]:
```
best_models =
```

RESULTS

Linear Regression:
Best Parameters: {}
Best Score: -4.299840000852746e+19
Lasso Regression:
Best Parameters: {'alpha': 0.01}
Best Score: 0.6817133978338602
Ridge Regression:
Best Parameters: {'alpha': 1}
Best Score: 0.6828938126728473
Decision Tree:
Best Parameters: {'max_depth': None, 'min_samples_split': 10}
Best Score: 0.9160457936334401
Random Forest:
Best Parameters: {'max_depth': 15, 'min_samples_split': 10, 'n_estimators': 50}
Best Score: 0.9138742812379153

Selcting Random Forest due to it being less prone to overfitting and being a better generalization model

In [68]:
```
model = best_models['Random Forest']['best_model']

model
```

Out[68]:
```
▾ RandomForestRegressor
RandomForestRegressor(max_depth=15, min_samples_split=10, n_estimators=50)
```

In [69]:
```
model.fit(X_train,y_train)
```

Out[69]:
```
▾ RandomForestRegressor
RandomForestRegressor(max_depth=15, min_samples_split=10, n_estimators=50)
```

In [70]:
```
model.score(X_test,y_test)
```

Out[70]:
```
0.9108886760138941
```

The model has 91% R2

In [72]:
```
import pickle
```

In [76]:
```
pickle.dump(model, open("model/model.pkl", "wb"))
```